

ADVANTAGES AND DISADVANTAGES OF SPACE EXPLORATION

Is exploring outer space worth the cost — and the risk? • LEVEL A



Space exploration has become one of the most debated topics of our time — inspiring millions, but dividing opinion on cost and priorities.

Over the decades, space exploration has become a major part of human history. From July 1969, when Neil Armstrong and Buzz Aldrin first landed on the moon, to the Mars Rover programmes and the images from the James Webb Space Telescope, exploring outer space continues to captivate millions of people around the world. Yet opinions are divided on whether space exploration is truly worthwhile. Is the enormous cost justified? And should we be spending that money on problems here on Earth instead?

Consider these two situations:

1 Your country wants to spend billions on a new space programme. What do you think?

a I think it's a waste of money — we have bigger problems to solve on Earth first.

b I think it's a good investment — space exploration leads to useful discoveries and new technology.

2 An astronaut's shuttle mission goes wrong and the crew is in danger. How does this affect your view?

a This proves space exploration is too dangerous — we should stop sending humans into space.

b Every field has risks — the information we gain from these missions helps us improve safety for the future.

These situations show the two sides of the space exploration debate. Both views are valid — and understanding the arguments for and against will help you decide which side you agree with.

■ ADVANTAGES

- Space research has led to life-changing **innovations** such as GPS, medical imaging, and new materials.
- It creates hundreds of thousands of jobs for scientists, engineers, and support staff.
- **Satellite** missions help scientists understand the Earth's **atmosphere** and predict natural disasters.
- Space exploration inspires young people to study science, technology, and engineering.
- For every dollar NASA spends, it generates roughly eight dollars within the economy.

■ DISADVANTAGES

- Space programmes cost enormous amounts of money — NASA's budget alone is over \$25 billion a year.
- Space exploration can be extremely dangerous — the Challenger **disaster** killed its entire **crew** of seven.
- Old satellites and equipment create space **debris** — NASA calls low Earth **orbit** an 'orbital graveyard'.
- Critics argue the money could be better spent on education, healthcare, and poverty on Earth.
- Most missions are not **cost-effective** and take decades to produce results.

\$25B+

NASA's annual budget in 2021

6,000T

of debris floating in low Earth orbit

x8

dollars generated per dollar NASA spends

1969

year humans first landed on the moon

The Dangers of Space Exploration

One of the most tragic accidents in the history of the American space programme was the **explosion** of the space **shuttle** Challenger. On 28 January 1986, the shuttle exploded just 73 seconds after launch, killing all seven **crew** members on board. The shuttle programme was suspended for nearly three years after the **disaster**. A second tragedy, the Columbia disaster, occurred in 2003 when the orbiter was destroyed upon re-entering Earth's **atmosphere**, resulting in another **tragic loss** of seven lives. These events showed the world that space travel carries real and serious risks.

However, supporters of space exploration argue that these disasters, while terrible, provided valuable lessons that helped **mitigate** danger in future missions. With greater experience comes better technology and improved safety — which is why many believe we should continue, not stop.

Innovations from Space Research

Even setting aside the disadvantages, research conducted to develop space programmes has led to inventions that benefit society in countless ways. GPS technology, breast cancer imaging, lightweight breathing systems for firefighters, and teflon-coated **fiberglass** used in modern buildings and stadiums all came from space research. **Satellite** missions also help scientists around the world study the Earth's **atmosphere**, track climate change, and improve our ability to predict natural disasters.

"The United States was not built by those who waited and rested. This country was conquered by those who moved forward — and so will space."

— President John F. Kennedy, 1962

ACTIVITIES

3

Read the article and answer True (T) or False (F).

- 1 The first moon landing took place in July 1969.
- 2 Space exploration has produced no practical benefits for everyday life.
- 3 The Challenger disaster happened 73 minutes after launch.
- 4 For every dollar NASA spends, it generates roughly eight dollars for the economy.
- 5 NASA has called low Earth orbit an 'orbital graveyard'.
- 6 GPS technology was developed as a result of space research.

4

Match the highlighted words and phrases to their meanings.

- 1 to hold someone's attention and interest completely
- 2 a vehicle used to carry astronauts into space and back
- 3 a very serious event that causes great harm or loss of life
- 4 to reduce the severity or danger of something
- 5 pieces of old equipment floating in space
- 6 new ideas or inventions that change the way we do things

5

Answer these questions in full sentences.

- 1 Why do some people think space exploration is a waste of money?
- 2 What happened during the Challenger disaster?
- 3 Give two examples of innovations that came from space research.
- 4 What do satellite missions help scientists understand?

6

Write a short paragraph (4–6 sentences) giving your own view on space exploration.

Use these phrases:

I think space exploration is ... / One advantage is ... / On the other hand ... / Overall, I believe ...

■ TALKING POINTS

Do you think exploring space is more important than solving problems on Earth? Why?

Would you ever want to travel to space? What would be the biggest challenge?

Should space exploration be funded by governments or private companies like SpaceX?

Is it worth risking human lives to explore space?